Name: ______________________

PROPORTION,
(PROPORTION IN CONTEXT)

Date: ________________

LO: I can solve real-world problems involving proportion including recipes, maps and scale.

Proportion in real life

Proportion in context means using **scaling** to solve everyday problems: adapting recipes, reading maps, converting currencies, or comparing rates.

Method: Find the value of one unit (unitary method), then multiply to find the amount needed.

Quick examples

●	Recipe for 4: use ¼ amounts for 1 person	unitary method
●	Map 1:50,000 means 1 cm = 500 m	scale conversion
●	£1 = €1.15; £80 = €92	currency conversion
●	1:25,000 means 1 cm = 25 km	1 cm = 25,000 cm = 250 m
	Double recipe for 4 i...	

Key idea

Find the value of 1 unit first (divide), then scale up to the amount needed (multiply).

For 1: total ÷ n | For m: × m

Unitary method: divide then multiply

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – scaling a recipe

A recipe for 6 people needs 300 g flour. How much for 10 people?

For 1 person: 300 ÷ 6 = 50 g

For 10 people: 50 × 10

= 500 g

$\boxed{\textbf{500 g}}$

Find the amount per person, then multiply by the new number.

Example 2 – map scales

A map has scale 1:25,000. Two towns are 8 cm apart on the map. Find the real distance.

1 cm = 25,000 cm

8 cm = 8 × 25,000 = 200,000 cm

200,000 cm = 2000 m = 2 km

$\boxed{\textbf{2 km}}$

Multiply the map distance by the scale factor, then convert units.

Example 3 – currency conversion

£1 = €1.12. Convert £65 to euros.

£1 = €1.12

£65 = 65 × 1.12

= €72.80

€72.80

Multiply the amount by the exchange rate.

Example 4 – best buy

Pack A: 500 g for £1.80. Pack B: 750 g for £2.55. Which is better value?

A: £1.80 ÷ 500 = 0.36p per gram

B: £2.55 ÷ 750 = 0.34p per gram

Pack B is cheaper per gram

Pack B is better value

Compare price per unit (per gram or per ml).

YOUR TURN – PRACTICE (deliberate practice)

1. Scaling recipes

A recipe for 4 people needs: 200 g butter, 150 g sugar, 3 eggs, 250 g flour. Find amounts for:

a) 2 people

b) 6 people

c) 10 people

d) 1 person

2. Map scales

A map has scale 1:50,000. Find the real distance for each map measurement.

a) 3 cm

b) 7 cm

c) 12 cm

d) 0.5 cm

3. Currency conversion

£1 = $1.25. Convert each amount.

a) £40 to dollars

b) £120 to dollars

c) $75 to pounds

d) $200 to pounds

4. Best value

Find the price per unit and state which is better value.

a) 500 ml for £1.20 or 750 ml for £1.65

b) 6-pack for £3.60 or 10-pack for £5.50

c) 400 g for £2.80 or 1 kg for £6.50

d) 2 litres for £1.50 or 3 litres for £2.10

5. Mixed proportion problems

Solve each real-world proportion problem.

a) A car uses 6 litres per 100 km. Fuel for 350 km?

b) Map 1:200,000. Real distance 15 km. Map distance?

c) Recipe for 8: 400 g. Amount for 5?

d) €1 = £0.87. Convert €50.

e) 3 painters take 8 hours. How long for 4 painters?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) The unitary method finds the value of 1 unit first ☐

Correct answer: __________

Reason: _________________

b) A scale of 1:50,000 means 1 cm represents 50 km ☐

Correct answer: __________

Reason: _________________

c) To compare value, find price per unit ☐

Correct answer: __________

Reason: _________________

d) Doubling a recipe for 4 gives enough for 6 ☐

Correct answer: __________

Reason: _________________

e) Map scale 1:25,000 means 1 cm = 250 m ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe for **12 cupcakes** uses 225 g flour, 175 g sugar and 200 ml milk. You want to make **30 cupcakes** but only have **500 g flour**. Is that enough? What is the maximum number you can make with 500 g flour?

Expression: ____________________________

Simplified: ____________________________

b) A model car is built to a scale of **1:18**. The real car is **4.5 m long** and **1.8 m wide**. Find the model dimensions. If the model weighs **1.2 kg**, can you determine the real car's weight from this information alone? Explain.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!